1. Brainstorming & Ideation Phase

Idea generation, problem identification and proposed solution

Purpose

This phase explains why EduGenie was proposed and what learning problem it aims to address.

Problem Identified

Students may need quick explanations, answers, summaries, quizzes and structured learning guidance without navigating many separate resources.

Project Idea

EduGenie is a lightweight AI-powered educational assistant designed to simplify learning through generative AI. The supplied project document identifies Q&A;, simplified explanations, quiz generation, personalized learning recommendations and passage summarization as core capabilities.

Target Users

Students, self-learners and learners at different academic levels.

Key Use Cases

- Ask an academic or general-knowledge question and receive an AI-generated answer.
- Enter a topic and request a simplified explanation.
- Provide a passage/topic and generate a quiz.
- Summarize a long educational passage for revision.
- Request a structured beginner-to-advanced learning path.

Proposed Technology Direction

FastAPI backend, HTML/CSS frontend, Google Gemini for cloud AI capabilities, and LaMini-Flan-T5 for lightweight local concept explanation.

Expected Outcome

A simple web-based learning assistant with multiple educational functions exposed through a common interface.